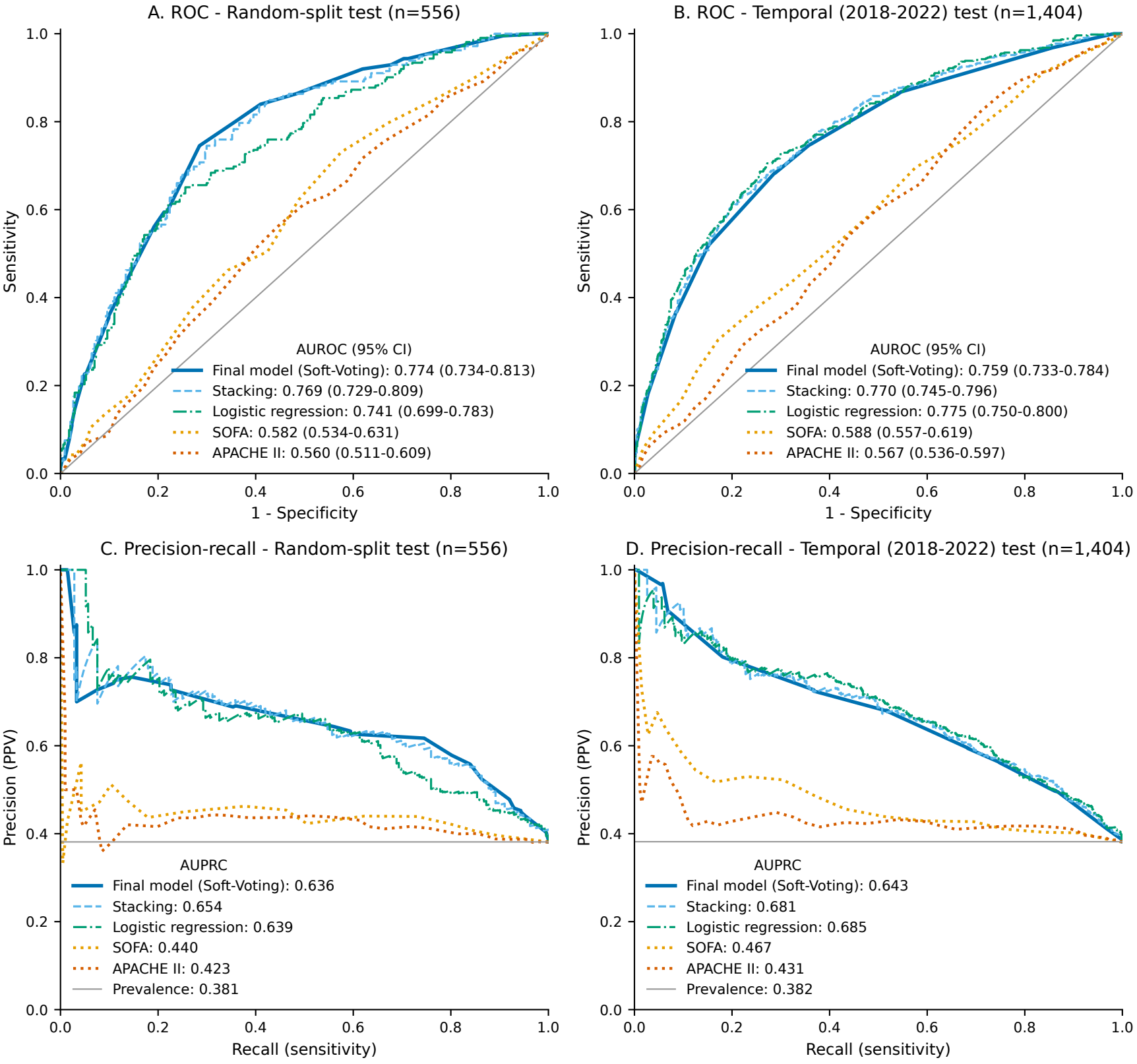
A. ROC - Random-split test (n=556)
Sensitivity
1 - Specificity
AUROC (95% CI)
Final model (Soft-Voting): 0.774 (0.734-0.813)
Stacking: 0.769 (0.729-0.809)
Logistic regression: 0.741 (0.699-0.783)
SOFA: 0.582 (0.534-0.631)
APACHE II: 0.560 (0.511-0.609)
B. ROC - Temporal (2018-2022) test (n=1,404)
Sensitivity
1 - Specificity
AUROC (95% CI)
Final model (Soft-Voting): 0.759 (0.733-0.784)
Stacking: 0.770 (0.745-0.796)
Logistic regression: 0.775 (0.750-0.800)
SOFA: 0.588 (0.557-0.619)
APACHE II: 0.567 (0.536-0.597)
C. Precision-recall - Random-split test (n=556)
Precision (PPV)
Recall (sensitivity)
AUPRC
Final model (Soft-Voting): 0.636
Stacking: 0.654
Logistic regression: 0.639
SOFA: 0.440
APACHE II: 0.423
Prevalence: 0.381
D. Precision-recall - Temporal (2018-2022) test (n=1,404)
Precision (PPV)
Recall (sensitivity)
AUPRC
Final model (Soft-Voting): 0.643
Stacking: 0.681
Logistic regression: 0.685
SOFA: 0.467
APACHE II: 0.431
Prevalence: 0.382